Limit Order Book Forecasting with Conditional Diffusion Models

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Abstract

Simulating limit order book (LOB) dynamics is central to market forecasting and evaluation. While diffusion models achieve strong empirical performance, existing LOB snapshot models typically lack a learned recurrent history encoder and formal statistical guarantees. We propose **LOBGrad**, a conditional DDPM that combines an autoregressive RNN encoder with a diffusion-based generator for probabilistic time-series forecasting. For stationary exponentially mixing transformed LOB sequences with bounded support, we establish an excess-risk oracle inequality for the DDPM noise-prediction objective. In particular, we show how the dependence on the conditioning window can be reduced to the latent-state dimension under contractive dynamics. Experiments on BTC/USDT data show that LOBGrad outperforms an encoder-free diffusion ablation, conditional Wasserstein GANs and statistical baselines on queue metrics, reproduces return distribution and autocorrelation characteristics, and yields reasonably calibrated predictive intervals for queue-derived quantities across a 20-step horizon, while return intervals remain less well calibrated.

1. Introduction

Limit order books (LOBs) are the main trading mechanism in modern electronic markets (Gould et al., 2013). A Level-2 LOB is an ordered collection of bid and ask queues that records outstanding buy and sell orders submitted by market participants, each specifying a price and an associated volume. The queue profile summarizes local liquidity, the supply of immediacy, and the short-horizon price impact faced by traders (Bouchaud et al., 2018). Accordingly, simulating LOB dynamics is central to tasks such as market forecasting, execution-strategy evaluation, and counterfactual market analysis (Backhouse et al., 2025; Wang & Ventre, 2025).

Denoising diffusion probabilistic models (DDPMs) (Ho et al., 2020; Song et al., 2021) have recently shown strong empirical performance in LOB simulation (Backhouse et al., 2025; Wang & Ventre, 2025). DDPMs can learn highly complex distributions (Chen et al., 2023b; Yu & Yu, 2026), making them a natural fit for modeling order book dynamics (Bouchaud et al., 2018). Building on these capabilities, DDPMs have also been extended to conditional time-series forecasting (Rasul et al., 2021; Chen et al., 2024). Despite this empirical success, existing diffusion-based LOB snapshot models typically do not explicitly use a learned recurrent history encoder, nor do they provide theoretical guarantees illustrating why a recurrent history encoder might enhance predictive performance.

In this work, we address this limitation by proposing **LOBGrad**, an encoder-conditioned denoising diffusion model for probabilistic LOB forecasting. Following the *TimeGrad* framework (Rasul et al., 2021), LOBGrad integrates an autoregressive RNN encoder with a diffusion-based conditional generator. The recurrent encoder maps past LOB states to a low-dimensional hidden state (Schäfer & Zimmermann, 2006; Sirignano & Cont, 2019; Hanson & Raginsky, 2020). This paper makes three contributions. First, we introduce a conditional diffusion architecture that jointly forecasts queue depths and mid-price returns from a recurrently encoded LOB history. Second, for stationary, exponentially mixing sequences, we establish the excess-risk bounds for diffusion models with a learned recurrent encoder, separating stochastic, denoising-head approximation, and encoder approximation errors. We show that under contractive latent-state dynamics, the encoder term depends on the hidden-state dimension m rather than on the flattened window dimension sd_y. Third, we show that LOBGrad outperforms baseline models across LOB-Bench metrics (Nagy et al., 2025) on BTC/USDT data[1], producing forecasts that match key microstructure stylized facts.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 introduces the LOB repre-

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[1]A `REPRODUCE.md` file with step-by-step instructions for reproduction is included with the submission.

sentation and our model architecture. Section 4 presents the theoretical results. Section 5 reports the experimental evaluation. Section 6 concludes the paper. We provide detailed proofs and experimental details in the Appendix.

2. Related Work

A growing body of work uses generative models for LOB simulation to reproduce realistic market dynamics. Early approaches employ GANs to generate order streams and learn conditional distributions of future LOB states (Coletta et al., 2021; Li et al., 2020). Notably, Cont et al. (2023) proposes a Wasserstein GAN that conditions on the current LOB state and demonstrates the ability to reproduce key stylized facts, establishing GANs as a competitive baseline for LOB simulation. More recent work has moved beyond GANs. Autoregressive state-space models tokenize message streams to generate LOB transitions (Nagy et al., 2023). TRADES uses a transformer-based diffusion model for order-level LOB generation and counterfactual analysis (Berti et al., 2025). Backhouse et al. (2025) recasts LOB snapshots as images with image-based diffusion. DiffLOB (Wang & Ventre, 2026) conditions on discrete market regime labels to support counterfactual queries. Despite their empirical success, none of the diffusion-based snapshot models above employs an autoregressive encoder. Their conditioning relies on current-state, spatial image structure, or regime-level information rather than on a learned sequential summary of the LOB history.

Theoretical analyses of diffusion models have mainly focused on unconditional distribution estimation (Chen et al., 2023a; Oko et al., 2023; Benton et al., 2024; Lee et al., 2023; Li & Yan, 2025). These results do not directly apply to LOBGrad, where the target law is conditional on a dependent LOB history and the noise predictor is not a standalone feedforward network but a composition of a recurrent encoder and a denoising head. Our result is complementary: for stationary, exponentially mixing sequences, we prove an excess-risk oracle inequality for the conditional DDPM ε-prediction objective. Under a contractive latent-state realization, the encoder term depends on the hidden-state dimension m rather than on the flattened window dimension sd_y.

3. Methodology

LOBGrad models the one-step conditional law of a transformed LOB state and obtains multi-step paths by autoregressive rollouts. This section defines the state representation, the conditional DDPM, and the empirical training objective used in the theory.

3.1. Limit Order Book Representation

Centered Order Book. Figure 1 provides an intuitive illustration of the format of LOB data. At time t, let a_t and b_t denote the best ask and the best bid, i.e., the lowest price at which there is positive sell volume and the highest price at which there is positive buy volume, respectively. Let $p_{\mathrm{mid}}(t) = (a_t + b_t)/2$. We work with Level-2 snapshots, where all orders at the same price are aggregated into a queue size. To remove the drift in absolute prices, we center the book at the mid-price rounded to the nearest tick, denoted by p_t, and retain d bid-side and d ask-side levels. For $j = 1, \dots, d$, $Q^-_{-j}(t)$ is the bid volume at price $p_t - j\delta$ stored with negative sign, and $Q^+_j(t)$ is the ask volume at price $p_t + (j-1)\delta$ stored with positive sign. Empty queues are set to zero. The centered queue vector is

$$X_t = \big(\underbrace{Q^-_{-d}(t), \dots, Q^-_{-1}(t)}_{\text{bid queues}}, \underbrace{Q^+_1(t), \dots, Q^+_d(t)}_{\text{ask queues}}\big) \in \mathbb{R}^{2d}. \tag{1}$$

The centered LOB state is $z_t = (p_t, X_t) \in \mathbb{R}^{d_y}$, with $d_y = 2d+1$. The price index p_t records where the local coordinate system is centered, while X_t records the local queue profile modeled by the diffusion network.

Preprocessing. For stability, we apply a componentwise signed-log transformation to the queue vector: $\psi(Q) = \mathrm{sign}(Q)\log(1 + |Q|)$, extending the log-volume compression used by Sirignano & Cont (2019) to our signed convention. We then clip the transformed queue features before training. These transformations compress the heavy-tailed queue-size distribution. The price levels p_t are mapped to returns $r_t = p_t - p_{t-1}$, which are then clipped to $[-r_{\max}, r_{\max}]$, with $r_{\max} = 12$ USDT in the main experiments. Throughout the main text and theory,

$$y_t = (r_t, \psi(X_t)) \in \mathbb{R}^{d_y}$$

denotes the transformed LOB state, consisting of the one-step price return and the transformed queue vector. Given a look-back length s, the observed history at time $t-1$ is

$$H_{t-1} = (y_{t-s}, \dots, y_{t-1}) \in (\mathbb{R}^{d_y})^s. \tag{2}$$

The forecasting target is the conditional distribution of y_t given H_{t-1}.

3.2. Encoder-Conditioned DDPM

Let $y_t^{(0)} = y_t$. For a variance schedule $\{\beta_k\}_{k=1}^K$, set $\alpha_k = 1 - \beta_k$ and $\bar\alpha_k = \prod_{j=1}^k \alpha_j$. The forward DDPM marginal is $y_t^{(k)} = \sqrt{\bar\alpha_k}\, y_t + \sqrt{1-\bar\alpha_k}\,\varepsilon_t$, $\varepsilon_t \sim \mathcal{N}(0, I_{d_y})$. The reverse sampler conditions on the observed history through a learned noise predictor $g^{(\mathrm{tg})}$, defined in (6). For reverse

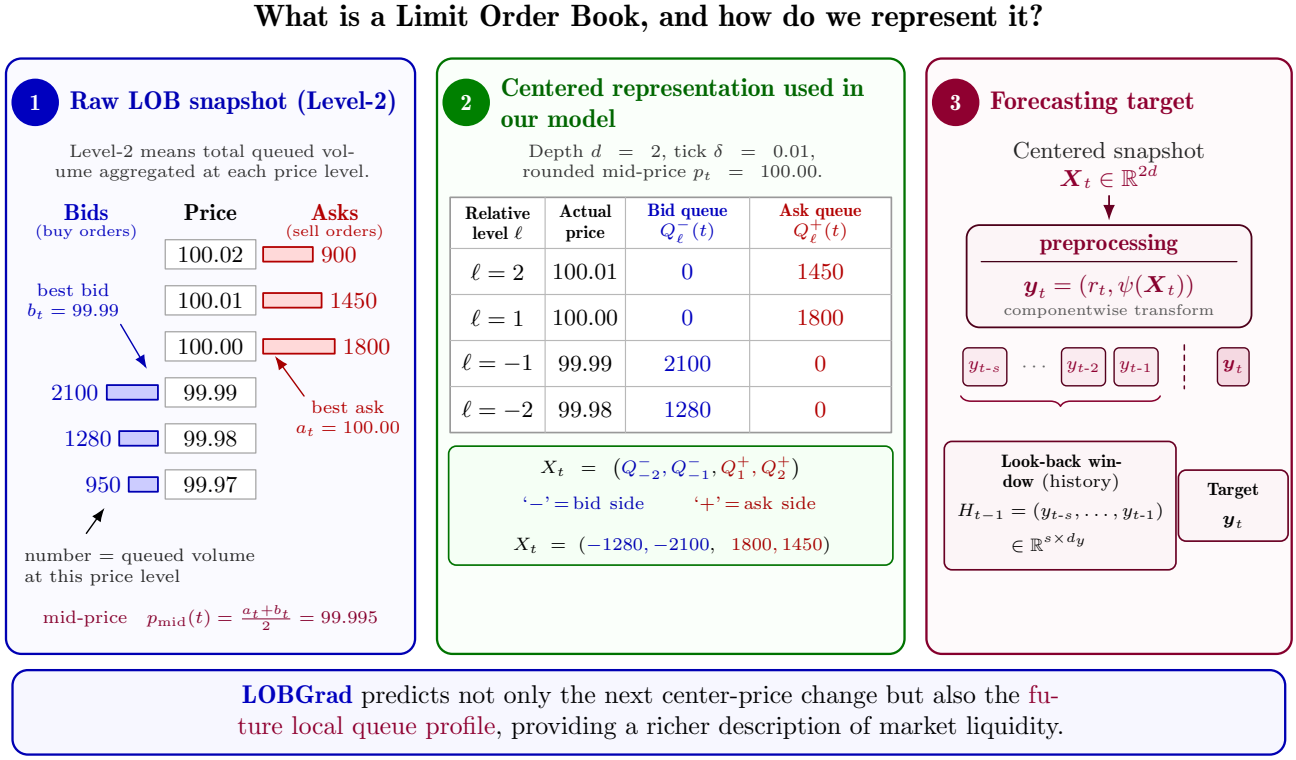


Figure 1. Illustration of the raw Level-2 limit order book, the centered representation used in our model, and the forecasting target.

step $k = K, \dots, 2$, we use

$$p(y_t^{(k-1)} \mid y_t^{(k)}, H_{t-1}) = \mathcal{N}\left(\frac{1}{\sqrt{\alpha_k}}\left[y_t^{(k)} - \frac{1-\alpha_k}{\sqrt{1-\bar{\alpha}_k}} g^{(\text{tg})}(H_{t-1}, y_t^{(k)}, k)\right], \beta_k I_{d_y}\right). \tag{3}$$

The learnable part of the conditional DDPM is the noise predictor $g^{(\text{tg})}$, which predicts the injected Gaussian noise from the noised target, the diffusion index, and the LOB history.

LOBGrad follows a TimeGrad-type principle of conditioning diffusion forecasts on an autoregressive summary of past observations (Rasul et al., 2021). The architecture has two components: a time-series encoder $g^{(\text{enc})}$ and a noise-prediction network $g^{(\epsilon)}$, which are defined in (4) and (5), respectively. Their composition defines the predictor $g^{(\text{tg})}$, whose statistical properties are analyzed in Section 4.

Time-Series Encoder. The encoder maps the look-back window to an m-dimensional conditioning state. We write

$$h_{t-1} = g^{(\text{enc})}(H_{t-1}) \in [-1,1]^m, \quad g^{(\text{enc})} \in \mathcal{G}^{(\text{enc})}_{V,L,\kappa}(s, m, d_y), \tag{4}$$

where h_{t-1} is the m-dimensional conditioning state summarizing the look-back window. LOBGrad realizes $g^{(\text{enc})}$ as a stacked GRU or LSTM (Appendix A). For the analysis of Section 4, we replace it by a clipped deep ReLU recurrent surrogate from $\mathcal{G}^{(\text{enc})}_{V,L,\kappa}(s, m, d_y)$ with the same input-output signature, width V, depth L, and parameter-norm bound κ, used only to obtain approximation and metric-entropy bounds.

Noise-Prediction Network. The noise-prediction network is an MLP. It takes the noised target $y_t^{(k)}$, the diffusion index k, and the history summary h_{t-1} as input:

$$g^{(\epsilon)} : \mathbb{R}^{d_y} \times \{1, \dots, K\} \times [-1,1]^m \longrightarrow \mathbb{R}^{d_y}, \quad g^{(\epsilon)} \in \mathcal{G}^{(\epsilon)}_{D,W,\kappa}(d_y, m, K), \tag{5}$$

where D denotes the depth of the network, W denotes its maximal hidden-layer width, and κ bounds the affine weights and biases in the network class.

Composite Function Class. The predictor used in the reverse DDPM is the clipped composition

$$g^{(\text{tg})}(H_{t-1}, y_t^{(k)}, k) := \Pi_{\text{clip}}\left(y_t^{(k)}, g^{(\epsilon)}\left(y_t^{(k)}, k, g^{(\text{enc})}(H_{t-1})\right)\right), \tag{6}$$

where $\Pi_{\text{clip}}(y, v)$ denotes coordinatewise clipping of v at level $C_{\text{clip}}(1 + \|y\|_\infty)$. Choosing C_{clip} sufficiently large leaves the oracle unchanged while controlling the empirical

process uniformly over the network class. The resulting network class is

$$\begin{aligned}&\mathcal{G}^{(\mathrm{tg})}_{V,L,D,W,\kappa}(s,m,d_y,K)\\&=\Big\{g^{(\mathrm{tg})}: g^{(\mathrm{enc})}\in\mathcal{G}^{(\mathrm{enc})}_{V,L,\kappa}(s,m,d_y),\\&\qquad g^{(\epsilon)}\in\mathcal{G}^{(\epsilon)}_{D,W,\kappa}(d_y,m,K)\Big\}.\end{aligned}\tag{7}$$

All dependence on the LOB history enters the diffusion sampler through the recurrent state h_{t-1}.

3.3. Training and Inference

Training Objective. From a single transformed LOB series, let T denote the number of usable training pairs after removing the first s observations. We reindex these overlapping pairs as $\{(H_{t-1},y_t)\}_{t=1}^{T}$. For each t, draw $N_t\sim\mathrm{Unif}\{1,\dots,K\}$ and $\varepsilon_t\sim\mathcal{N}(0,I_{d_y})$, independently of the LOB process, and form $y_t^{(N_t)}$. For any network $g\in\mathcal{G}^{(\mathrm{tg})}_{V,L,D,W,\kappa}$, define empirical and population denoising risks as

$$\begin{aligned}\mathcal{R}_T(g)&:=\frac{1}{T}\sum_{t=1}^{T}\Big\|\varepsilon_t-g(H_{t-1},y_t^{(N_t)},N_t)\Big\|_2^2,\\\mathcal{R}(g)&:=\mathbb{E}\left[\Big\|\varepsilon_t-g(H_{t-1},y_t^{(N_t)},N_t)\Big\|_2^2\right].\end{aligned}\tag{8}$$

LOBGrad estimates the conditional denoiser by empirical risk minimization

$$\widehat{g}_T^{(\mathrm{tg})}\in\operatorname{argmin}_{g\in\mathcal{G}^{(\mathrm{tg})}_{V,L,D,W,\kappa}(s,m,d_y,K)}\mathcal{R}_T(g).\tag{9}$$

In practice, (9) is optimized by stochastic gradient descent, with one Monte Carlo draw of (N_t,ε_t) per training example and update. Section 4 studies the statistical error of the global empirical minimizer $\widehat{g}_T^{(\mathrm{tg})}$. Optimization error from SGD is not included in the theory.

Inference. Given a new history H_{t-1}, we generate a one-step forecast by initializing $y_t^{(K)}\sim\mathcal{N}(0,I_{d_y})$ and iterating the reverse update for $k=K,\dots,1$ according to

$$\begin{aligned}y_t^{(k-1)}=&\tfrac{1}{\sqrt{\alpha_k}}\left[y_t^{(k)}-\tfrac{1-\alpha_k}{\sqrt{1-\bar\alpha_k}}\,\widehat{g}_T^{(\mathrm{tg})}(H_{t-1},y_t^{(k)},k)\right]\\&+\sqrt{\beta_k}\,z_k,\end{aligned}\tag{10}$$

where $z_k\sim\mathcal{N}(0,I_{d_y})$ for $k>1$ and $z_1=0$. The terminal value $y_t^{(0)}$ is a draw from the learned conditional distribution of the next transformed LOB state. Multi-step paths are obtained by appending generated states to the history window and repeating the same one-step sampler autoregressively. To recover a raw LOB snapshot, we apply the inverse preprocessing: the first component is the predicted one-step mid-price return, which can be accumulated against a stored anchor price p_{t-1} to reconstruct the absolute price series, while each queue component v is mapped back to raw volume via ψ^{-1}.

4. Theoretical Guarantees

Notation. Throughout this section, T denotes the sample size, $\|\cdot\|_\infty$ the entrywise sup-norm, and $\|\cdot\|_2$ the Euclidean norm. We write $a_T\lesssim b_T$ if $a_T\le C\,b_T$ for some absolute constant $C>0$, and $a_T\asymp b_T$ if $a_T\lesssim b_T\lesssim a_T$. The notations $\widetilde{O}(\cdot)$ and $\widetilde{O}_{\mathbb{P}}(\cdot)$ denote the standard $O(\cdot)$ and $O_{\mathbb{P}}(\cdot)$ orders up to factors polynomial in $\log T$.

We establish the statistical guarantees for the DDPM ε-prediction risk in this section. Let

$$g^\star(H,Y^{(k)},k)=\mathbb{E}\Big[\varepsilon\mid H,Y^{(k)},k\Big]\tag{11}$$

be the oracle noise predictor for the DDPM noise-prediction objective. For any measurable g with finite risk, the excess risk is $\mathcal{R}(g)-\mathcal{R}(g^\star)$. By the projection identity for squared loss,

$$\begin{aligned}&\mathcal{R}(g)-\mathcal{R}(g^\star)\\&=\mathbb{E}\left[\Big\|g(H,Y^{(N)},N)-g^\star(H,Y^{(N)},N)\Big\|_2^2\right].\end{aligned}\tag{12}$$

Thus, consistency for the DDPM objective is equivalent to estimating the conditional mean of the injected noise.

Assumption 4.1 (Statistical structure)**.** The transformed LOB process $(y_t)_{t\in\mathbb{Z}}$ is stationary and exponentially α-mixing. There exists a constant $B_y<\infty$ such that $\|y_t\|_\infty\le B_y$ almost surely. Let $H=(y_{t-s},\dots,y_{t-1})$ be the history. There exist an m-dimensional summary map $\Gamma^\dagger:([-B_y,B_y]^{d_y})^s\to[-1,1]^m$ and a reduced oracle map $\bar g^\star:\mathbb{R}^{d_y}\times\{1,\dots,K\}\times[-1,1]^m\to\mathbb{R}^{d_y}$ such that the oracle noise predictor factors as

$$g^\star(H,Y^{(k)},k)=\bar g^\star\Big(Y^{(k)},k,\Gamma^\dagger(H)\Big).\tag{13}$$

The map $\Gamma^\dagger$ is γ-Hölder on $([-B_y,B_y]^{d_y})^s$. For each k, $\bar g^\star(\cdot,k,\cdot)$ is β-Hölder on every compact set $[-B_y-r,B_y+r]^{d_y}\times[-1,1]^m$, with Hölder radius at most polynomial in r.

Remark 4.2. The map $\Gamma^\dagger$ is an ideal population encoder. It is the low-dimensional state of the history that is sufficient for the one-step DDPM noise-prediction. The factorization (13) means that, for the purpose of predicting the injected noise from $(H,Y^{(k)},k)$, the full history H enters the oracle only through the summary $h=\Gamma^\dagger(H)$. The map $\bar g^\star$ is the corresponding oracle noise predictor function written in the reduced variables $(Y^{(k)},k,h)$. Note that the bar does not denote averaging. The normalization $\Gamma^\dagger(H)\in[-1,1]^m$ is not substantive and can be replaced by any bounded range

with only fixed changes in constants. For $0 < a \le 1$, an a-Hölder map f satisfies $\|f(x) - f(x')\|_\infty \le L\|x - x'\|_\infty^a$. For $a > 1$, the derivatives up to order $\lfloor a \rfloor$ are bounded, and the derivatives of order $\lfloor a \rfloor$ are $(a - \lfloor a \rfloor)$-Hölder. Thus, β and γ measure the smoothness of the functions $\bar{g}^\star$ and $\Gamma^\dagger$, respectively. The remaining proof-level regularity conditions, including the entropy and envelope bounds for the clipped composite class, are stated and verified in Appendix C.

In the proof, we localize the Gaussian perturbation to $r_T \asymp \sqrt{\log T}$, and the resulting logarithmic factors are absorbed into $\widetilde{O}_\mathbb{P}$. Define

$$
\begin{aligned}
&A^{\mathrm{head}}(D, W) \qquad (14)\\
:= &\inf_{g^{(\epsilon)} \in \mathcal{G}^{(\epsilon)}_{D,W,\kappa}(d_y, m, K)} \max_{1 \le k \le K} \left\| g^{(\epsilon)}(\cdot, k, \cdot) - \bar{g}^\star(\cdot, k, \cdot) \right\|^2_{\infty, \Xi_{r_T}},\\
&\Delta_{\mathrm{enc}}(V, L) \qquad (15)\\
:= &\inf_{g^{(\mathrm{enc})} \in \mathcal{G}^{(\mathrm{enc})}_{V,L,\kappa}(s, m, d_y)} \sup_{H \in ([-B_y, B_y]^{d_y})^s} \left\| g^{(\mathrm{enc})}(H) - \Gamma^\dagger(H) \right\|_\infty,
\end{aligned}
$$

where $\Xi_r = [-B_y - r, B_y + r]^{d_y} \times [-1, 1]^m$. Let $\beta_0 = \min\{\beta, 1\}$ and write the metric-entropy complexity of the clipped composite class as

$$
\begin{aligned}
\mathfrak{S}(V, L, D, W) := &\big((2s-1)V + W + d_y + 1\big)^2\\
&\times (sL + D + 2)^2\\
&\times \log\big((V + W)(L + D + 2)\big).
\end{aligned} \qquad (16)
$$

The complexity is verified in Appendix C.4.

Theorem 4.3 (Excess risk for encoder-conditioned DDPMs)**.** *Suppose Assumption 4.1 holds and $\widehat{g}^{(\mathrm{tg})}_T$ is the empirical minimizer in (9). Then*

$$
\begin{aligned}
&\mathcal{R}(\widehat{g}^{(\mathrm{tg})}_T) - \mathcal{R}(g^\star)\\
=&\widetilde{O}_\mathbb{P}\left(\sqrt{\frac{\mathfrak{S}(V, L, D, W)}{T}} + \frac{\mathfrak{S}(V, L, D, W)}{T} \right.\\
&\left. + A^{\mathrm{head}}(D, W) + \Delta_{\mathrm{enc}}(V, L)^{2\beta_0} \right).
\end{aligned} \qquad (17)
$$

For fixed recurrent depth L and noise-prediction network depth $D = O(\log W)$, $\mathfrak{S}(V, L, D, W) = \widetilde{O}(V^2 + W^2)$.

Remark 4.4. The first two terms in (17) are the stochastic cost of empirical risk minimization under dependence. The last two terms are approximation errors: A^{head} is the error of the denoising MLP when the ideal summary is supplied, while $\Delta_{\mathrm{enc}}^{2\beta_0}$ is the additional error from replacing $\Gamma^\dagger(H)$ by a learned recurrent summary.

Standard ReLU approximation bounds give, up to logarithmic factors,

$$
\begin{aligned}
A^{\mathrm{head}}(D, W) &\lesssim W^{-2\beta/(d_y+m)},\\
\Delta_{\mathrm{enc}}(V, L_0)^{2\beta_0} &\lesssim V^{-4\gamma\beta_0/(sd_y)},
\end{aligned} \qquad (18)
$$

for fixed encoder depth L_0 and head depth schedule $D = O(\log W)$.

Corollary 4.5 (Generic window rate)**.** *Under Assumption 4.1, choose*

$$
W \asymp T^{\frac{d_y+m}{2(d_y+m+2\beta)}}, \qquad V \asymp T^{\frac{sd_y}{2(sd_y+4\gamma\beta_0)}}. \qquad (19)
$$

Then

$$
\mathcal{R}(\widehat{g}^{(\mathrm{tg})}_T) - \mathcal{R}(g^\star) = \widetilde{O}_\mathbb{P}\left(T^{-\frac{\beta}{d_y+m+2\beta}} + T^{-\frac{2\gamma\beta_0}{sd_y+4\gamma\beta_0}} \right). \qquad (20)
$$

Corollary 4.5 treats H as a flattened vector in $\mathbb{R}^{sd_y}$, so the encoder term deteriorates with the look-back dimension. A sharper bound is available when the sufficient summary is generated by a stable m-dimensional recurrence. Specifically, let $H = (u_1, \dots, u_s)$ and suppose

$$
\begin{aligned}
&h^\dagger_0 = 0,\\
&h^\dagger_j = \Phi^\dagger(P^\dagger u_j + h^\dagger_{j-1}), \quad j = 1, \dots, s,\\
&\Gamma^\dagger(H) = h^\dagger_s,
\end{aligned} \qquad (21)
$$

where $P^\dagger : \mathbb{R}^{d_y} \to \mathbb{R}^m$ is a fixed lift, $\Phi^\dagger$ is locally γ-Hölder, and $\|\Phi^\dagger(P^\dagger u + h) - \Phi^\dagger(P^\dagger u + h')\|_\infty \le \lambda \|h - h'\|_\infty$ for some $\lambda < 1$.

Corollary 4.6 (Hidden-state refinement)**.** *Under (21), estimate over the corresponding state-space recurrent subclass and let $\widehat{g}^{(\mathrm{tg})}_{T,\mathrm{st}}$ be the resulting ERM. Let Δ_{st} denote the analogue of Δ_{enc} over this subclass. The encoder approximation term becomes*

$$
\Delta_{\mathrm{st}}(V, L_1)^{2\beta_0} \lesssim V^{-4\gamma\beta_0/m}, \qquad (22)
$$

up to logarithmic factors. With

$$
W \asymp T^{\frac{d_y+m}{2(d_y+m+2\beta)}}, \qquad V \asymp T^{\frac{m}{2(m+4\gamma\beta_0)}},
$$

we obtain

$$
\mathcal{R}(\widehat{g}^{(\mathrm{tg})}_{T,\mathrm{st}}) - \mathcal{R}(g^\star) = \widetilde{O}_\mathbb{P}\left(T^{-\frac{\beta}{d_y+m+2\beta}} + T^{-\frac{2\gamma\beta_0}{m+4\gamma\beta_0}} \right). \qquad (23)
$$

The refinement explains the statistical role of the recurrent encoder: under stable latent-state dynamics, the statistical cost of using a long look-back window is governed by the hidden-state dimension m rather than by the flattened dimension sd_y. The guarantee is for the one-step DDPM ε-prediction risk. Multi-step LOB paths are obtained by autoregressive rollout and are evaluated empirically in Section 5.

5. Experimental Results

We evaluate LOBGrad on BTC/USDT Level-2 limit order book snapshots collected from the Binance public REST depth endpoint with depth 100 on each side, covering October 1–21, 2023. The raw data are obtained from asynchronous polling logs. We construct the working sample by retaining snapshots at least 5000 ms apart and discarding malformed records or snapshots with an empty bid or ask side, without interpolation or forward filling. This yields 308,826 snapshots, with median inter-snapshot spacing 6.17 seconds. The dataset is split chronologically into 15 training days, 2 validation days, and 4 test days. Each snapshot is represented by a centered signed queue vector with $d = 5$ levels per side and grid spacing $\delta = 2$ USDT, giving $d_y = 2d + 1 = 11$ in (1). Figure 5 in Appendix B shows that the choice of d and δ covers approximately 98% of the observed mid-price changes between consecutive snapshots.

We compare LOBGrad against three ablations (LG-Q, LG-R, and LG-NE) and three external baselines, including a conditional Wasserstein GAN (CGAN) (Cont et al., 2023), a multivariate Hawkes process (Jain et al., 2024), and an independent Poisson zero-intelligence model (Cont et al., 2010). LG-Q and LG-R use queue-only and return-only targets, respectively, while LG-NE removes the recurrent encoder and conditions on the flattened history of dimension $sd_y = 1100$, partially isolating the effect of recurrent summarization. This model is the proxy of previous diffusion-based LOB models in the literature (Backhouse et al., 2025; Wang & Ventre, 2026). For learnable baselines, hyperparameters are selected on the validation split using a common Optuna protocol with median pruning. Non-neural baselines are calibrated on the same preprocessed snapshot series. Table 5 in Appendix B reports the selected configuration, including look-back length $s = 100$, hidden size $m = 151$, and $K = 102$ diffusion steps. The empirical study covers three components: distributional generation quality on 500 held-out windows with 100-step contexts, 20-step forecast horizons, and 100 autoregressive rollouts per window; stylized-fact recovery for returns and queue dynamics; and a downstream limit-order execution task. All experiments are conducted on a single RTX 5080 GPU. Implementation details are deferred to Appendix B.

5.1. Generation Quality

Following LOB-Bench (Nagy et al., 2025), we report Wasserstein-1 distances for best-level total queue volume, order-book imbalance, and the marginal return distribution. We complement these with queue-profile MAE, queue and absolute-return ACF MAE, CRPS, and calibration coverage rates. Detailed definitions are given in Appendix B.3.

Table 1 shows that LOBGrad achieves the best score on four of the five queue metrics. The only exception is Queue ACF MAE, where the queue-only ablation LG-Q is marginally better, suggesting a small trade-off between modeling queue–return dependence and preserving short-lag queue autocorrelation. On return metrics, LG-R performs best on return Wasserstein and return CRPS, as expected from its dedicated return-only target. Nevertheless, LOBGrad improves substantially over the encoder-free diffusion baseline LG-NE on all queue metrics and on return distributional scores. Since LG-NE keeps the diffusion objective fixed but removes the recurrent summary $g^{(\mathrm{enc})}(H_{t-1})$, the gap in queue profile, ACF, and CRPS supports the empirical value of recurrent summarization. This observation is consistent with Corollary 4.6. However, note that this is not a formal test of the asymptotic bound, since Table 1 reports rollout metrics rather than the population DDPM ε-risk.

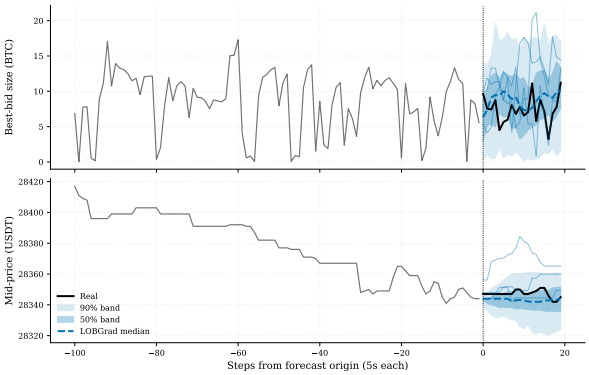


Figure 2. Sample best-bid queue (top) and mid-price (bottom) trajectories from LOBGrad with 50% and 90% predictive intervals on a held-out test window.

Figure 2 illustrates the same conclusion at the path level. The predictive bands from LOBGrad cover the realized best-bid queue and mid-price trajectory in a held-out window, while the CRPS curves in Figure 3 show that LOBGrad maintains the lowest queue CRPS throughout the 20-step forecast horizon and remains competitive with LG-R on return CRPS. Appendix Table 4 further shows that LOBGrad attains the lowest queue CRPS across volatility regimes. In addition to CRPS, Table 2 reports direct interval-calibration diagnostics with $M = 500$ rollouts. The 90% intervals are close to nominal for queue-derived quantities, with empirical coverages 0.876 for all queue dimensions, 0.892 for best-level volume, and 0.900 for order-book imbalance. Return coverage is lower at 0.844, indicating that the model captures queue uncertainty more reliably than extreme return variation.

5.2. Stylized Facts

Figure 4 evaluates whether LOBGrad reproduces standard price-series stylized facts. We compare real and generated successive-snapshot mid-price changes using the marginal

Table 1. Distributional metrics on τ=20-step horizon across 500 held-out windows. Lower is better. **Bold** marks the best value per row. Sub-rows give 95% bootstrap CIs over the evaluation windows (1000 resamples). NE = no encoder; Q = queue only; R = return only. Dashes indicate metrics that are not applicable

Metric	LOBGrad	LG-NE	LG-Q	LG-R	CGAN	Hawkes	Poisson
Queue							
L1 volume Wasserstein	**0.100** [0.084, 0.116]	0.104 [0.090, 0.119]	0.135 [0.112, 0.158]	—	0.197 [0.185, 0.209]	0.266 [0.242, 0.290]	0.330 [0.313, 0.347]
OB imbalance Wasserstein	**0.095** [0.071, 0.119]	0.107 [0.081, 0.134]	0.096 [0.071, 0.121]	—	0.196 [0.178, 0.215]	0.219 [0.197, 0.242]	0.280 [0.264, 0.296]
Queue profile MAE	**0.037** [0.035, 0.049]	0.109 [0.101, 0.120]	0.049 [0.047, 0.060]	—	0.044 [0.042, 0.055]	0.098 [0.092, 0.110]	0.100 [0.094, 0.110]
Queue ACF MAE	0.026 [0.018, 0.038]	0.081 [0.069, 0.093]	**0.024** [0.015, 0.036]	—	0.034 [0.025, 0.044]	0.225 [0.214, 0.238]	0.219 [0.209, 0.232]
CRPS (queue)	**0.261** [0.254, 0.268]	0.276 [0.268, 0.284]	0.262 [0.255, 0.268]	—	0.284 [0.276, 0.291]	0.341 [0.332, 0.350]	0.312 [0.304, 0.321]
Returns							
Return Wasserstein	0.097 [0.077, 0.118]	0.211 [0.187, 0.236]	—	**0.085** [0.077, 0.098]	0.280 [0.266, 0.294]	—	—
Absolute returns ACF MAE	0.011 [0.006, 0.019]	**0.008** [0.005, 0.017]	—	**0.008** [0.005, 0.017]	0.017 [0.011, 0.025]	—	—
CRPS (returns)	1.719 [1.576, 1.864]	1.760 [1.613, 1.909]	—	**1.694** [1.557, 1.836]	1.837 [1.681, 1.995]	—	—

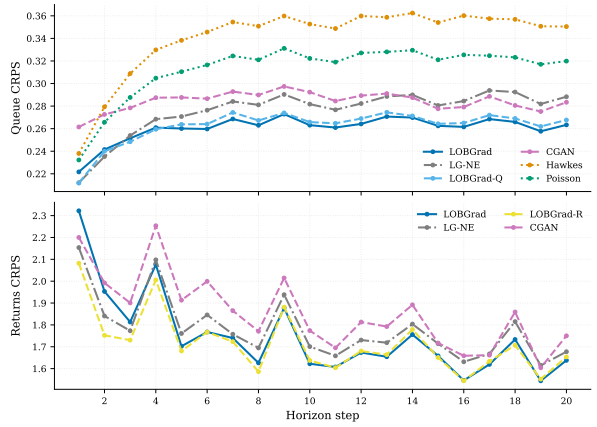


Figure 3. Per-step queue (top) and returns (bottom) CRPS over the 20-step generation horizon. LOBGrad maintains low queue CRPS throughout the forecast horizon.

return distribution, the return ACF, and the absolute-return ACF. Generated samples are obtained from 100 autoregressive rollouts for each of the 500 evaluation windows. LOBGrad closely matches the empirical return distribution, captures the small first-lag autocorrelation in returns, and partially recovers the short-horizon dependence in absolute returns. Additional queue diagnostics in Appendix B show that LOBGrad also reproduces the cross-level correlation structure more accurately than CGAN, with queue-correlation RMSE 0.079 versus 0.108, and closely matches the bid and ask depth profiles.

Table 2. Calibration summary on BTC/USDT test windows using $M = 500$ rollouts per window and horizon $\tau = 20$. Reported are empirical coverages for nominal 50% and 90% central intervals, and the mean width of the 90% interval.

Target	50% cov.	90% cov.	Mean 90% width
Returns	0.630	0.844	8.684
Queues ($2d = 10$)	0.442	0.876	6.213
Best-level volume	0.483	0.892	11.034
OB imbalance	0.479	0.900	1.243

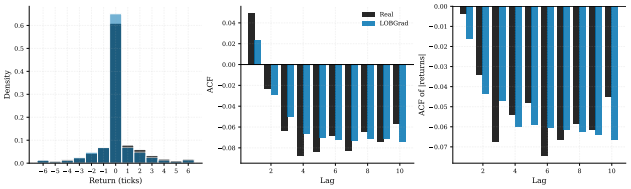


Figure 4. Stylized facts of 5-second mid-price returns: real held-out data (black) versus LOBGrad samples (blue). Return distribution (left), ACF of returns (middle) and ACF of absolute returns (right).

5.3. Downstream Task

Beyond distributional metrics, we test whether forecast quality supports useful execution decisions. We consider a stylized single-shot sell problem. At the start of each test window, an agent must sell a quantity Q of BTC. The order size Q is fixed across windows within a run and calibrated from training-set bid-side top-of-book depth statistics. The action $j = -1$ denotes an immediate market sell, while

$j \geq 0$ posts a passive limit-sell order at $p_{\text{lim}} = p_{\text{mid}}(0) + j\delta$, where $\delta = 2$ USDT, and waits up to 20 future snapshots. We consider offsets $j \in \{-1, 0, 1, 2, 3\}$. The order is treated as passively filled if the mid-price reaches the limit price or if the queue at the corresponding ask level is sufficiently depleted. Otherwise, it is executed at the final snapshot by walking the bid book. This setup creates the intended price–fill trade-off: small offsets fill more reliably but at less favorable prices, while larger offsets improve passive-price opportunities but increase adverse force-out risk. For each window, LOBGrad draws 100 autoregressive rollouts, simulates each candidate offset on each rollout, and selects the offset with the lowest mean simulated implementation shortfall $\text{IS} = p_{\text{mid}}(0) - \bar{p}_{\text{fill}}$, where $\bar{p}_{\text{fill}}$ is the realized average fill price. We report mean IS and $\text{CVaR}_{95}(\text{IS})$, the average shortfall over the worst 5% of windows. The strategies for evaluation are: (i) static rules with fixed j; (ii) LOBGrad-optimized selection, which draws 100 autoregressive rollouts, simulates all candidate offsets, and chooses the offset with lowest mean simulated IS; and (iii) an Oracle policy that uses the realized future path and is therefore an ex-post, non-implementable lower bound.

Table 3 shows the mean–tail trade-off in this execution problem. The immediate market sell is always completed and therefore has the lowest non-Oracle tail risk, but it pays the spread and has poor mean IS. Passive static offsets improve the mean when filled, but expose the agent to costly terminal market execution when unfilled. Among static rules, $j = 2$ attains the lowest mean IS at 0.56 USDT. LOBGrad-optimized further reduces mean IS to 0.03 USDT and also lowers CVaR_{95} relative to the best-mean static rule, from 47.86 to 43.22 USDT. This indicates that the generated rollouts contain useful information for selecting between aggressive and passive execution in individual windows, although the market action remains preferable under a purely tail-risk objective.

Table 3. Limit-order execution benchmark across 500 test windows. IS per share in USDT (lower is better). 95% bootstrap CIs computed by resampling windows with replacement 10,000 times. **Bold**: best non-Oracle row per column.

Strategy	Mean IS	$\text{CVaR}_{95}(\text{IS})$
$j = -1$ (Market)	5.89 [5.74, 6.04]	**8.45** [8.34, 8.53]
$j = 0$	1.75 [0.94, 2.65]	38.90 [29.43, 47.52]
$j = 1$	0.75 [−0.18, 1.78]	43.42 [34.11, 51.49]
$j = 2$	0.56 [−0.61, 1.77]	47.86 [38.37, 56.99]
$j = 3$	2.93 [1.60, 4.33]	51.51 [42.56, 59.76]
LOBGrad-optimized	**0.03** [−1.03, 1.18]	43.22 [35.39, 50.13]
Oracle	-6.73 [−7.14, −6.32]	7.07 [2.99, 7.74]

6. Conclusion

In this paper, we introduce LOBGrad, an encoder-conditioned denoising diffusion model for probabilistic LOB forecasting. The model couples a recurrent history encoder with a conditional noise predictor and generates joint queue–return paths by autoregressive rollout. We establish the excess-risk bounds for the DDPM ε-prediction objective under stationary exponentially mixing LOB sequences with bounded transformed states, separating stochastic error, denoising-head approximation, and encoder approximation. The hidden-state refinement shows that, under a contractive latent-state realization, the history-related term depends on the latent dimension m rather than the flattened window dimension sd_y. Empirically, LOBGrad outperforms GAN and statistical baselines on queue distributional metrics, generates multi-step forecasts that reproduce key microstructure stylized facts, and attains the lowest mean implementation shortfall on a downstream execution-risk task.

Several limitations remain. The theory controls the one-step DDPM ε-prediction risk of a global empirical risk minimizer under bounded transformed states, exponential mixing, and a low-dimensional contractive latent-state representation. It does not cover SGD optimization error, reverse-sampler distributional error, or error accumulation in autoregressive rollouts. Empirically, the study is limited to one BTC/USDT sample period and a stylized offline execution task. Future work includes deriving distributional consistency and applying LOBGrad to other assets, markets, and sampling regimes.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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Technical Appendices and Supplementary Material

A. Recurrent encoder architecture

The time-series encoder $g^{(\mathrm{enc})}$ introduced in Section 3 is, in practice, a stacked GRU or LSTM. This appendix gives the precise definition of the corresponding function classes $\mathcal{G}^{(\mathrm{GRU})}_{L,\kappa}(s,m,d_y)$ and $\mathcal{G}^{(\mathrm{LSTM})}_{L,\kappa}(s,m,d_y)$. Both share the input-output signature $\mathbb{R}^{s\times d_y}\to[-1,1]^m$ used by the deep ReLU recurrent surrogate $\mathcal{G}^{(\mathrm{enc})}_{V,L,\kappa}(s,m,d_y)$ of Section 4; in the GRU/LSTM realization the surrogate width V collapses to the hidden size m.

A.1. GRU

Throughout, fix an input dimension $d_y\in\mathbb{N}$, a hidden size $m\in\mathbb{N}$, a number of layers $L\in\mathbb{N}$, a look-back length $s\in\mathbb{N}$, and a parameter-norm bound $\kappa>0$. The encoder takes a sequence

$$H_{t-1}=(y_{t-s},\dots,y_{t-1})\in\mathbb{R}^{s\times d_y} \tag{24}$$

and returns the conditioning state

$$h_{t-1}\;:=\;h_{L,t-1}\;\in\;[-1,1]^m, \tag{25}$$

i.e., the top-layer final hidden state. Let $\sigma(u)=(1+e^{-u})^{-1}$ denote the element-wise sigmoid and $\odot$ the Hadamard product. Initial hidden states are taken to be zero.

For each layer $\ell\in\{1,\dots,L\}$ and each time index $i\in\{t-s,\dots,t-1\}$, the input to layer ℓ at step i is

$$x_{\ell,i}=\begin{cases}y_i, & \ell=1,\\ h_{\ell-1,i}, & 2\le\ell\le L,\end{cases} \tag{26}$$

so that the encoder is the deterministic map obtained by composing the recursions below across layers and time.

We define the first GRU layer class $\mathcal{G}^{(\mathrm{GRU})}_{1,\kappa}(m,d_y)$ as the set of maps $g_1^{(\mathrm{GRU})}:\mathbb{R}^{s\times d_y}\to\mathbb{R}^{s\times m}$, and for $\ell\in\{2,\dots,L\}$ we define the ℓ-th GRU layer class $\mathcal{G}^{(\mathrm{GRU})}_{\ell,\kappa}(m)$ as the set of maps $g_\ell^{(\mathrm{GRU})}:\mathbb{R}^{s\times m}\to\mathbb{R}^{s\times m}$, whose hidden-state recursion, for $i=t-s,\dots,t-1$, reads

$$r_{\ell,i}=\sigma\left(W_\ell^{(r)}x_{\ell,i}+U_\ell^{(r)}h_{\ell,i-1}+b_\ell^{(r)}\right), \tag{27}$$

$$z_{\ell,i}=\sigma\left(W_\ell^{(z)}x_{\ell,i}+U_\ell^{(z)}h_{\ell,i-1}+b_\ell^{(z)}\right), \tag{28}$$

$$\widetilde{h}_{\ell,i}=\tanh\left(W_\ell^{(h)}x_{\ell,i}+r_{\ell,i}\odot\left(U_\ell^{(h)}h_{\ell,i-1}\right)+b_\ell^{(h)}\right), \tag{29}$$

$$h_{\ell,i}=\left(1-z_{\ell,i}\right)\odot\widetilde{h}_{\ell,i}+z_{\ell,i}\odot h_{\ell,i-1}. \tag{30}$$

Here $r_{\ell,i},z_{\ell,i}\in(0,1)^m$ are the reset and update gates, and $\widetilde{h}_{\ell,i}\in(-1,1)^m$ is the candidate hidden state. The input weight matrices are $W_1^{(r)},W_1^{(z)},W_1^{(h)}\in\mathbb{R}^{m\times d_y}$ for the first layer and $W_\ell^{(r)},W_\ell^{(z)},W_\ell^{(h)}\in\mathbb{R}^{m\times m}$ for $\ell\in\{2,\dots,L\}$; the recurrent weight matrices are $U_\ell^{(r)},U_\ell^{(z)},U_\ell^{(h)}\in\mathbb{R}^{m\times m}$; and the bias vectors of the reset gate, update gate, and candidate hidden state are $b_\ell^{(r)},b_\ell^{(z)},b_\ell^{(h)}\in\mathbb{R}^m$. All entries of these weight matrices and bias vectors are bounded in absolute value by κ. Initial hidden states are taken to be zero, $h_{\ell,t-s-1}=0$.

We define the GRU encoder function class $\mathcal{G}^{(\mathrm{GRU})}_{L,\kappa}(s,m,d_y)$ as the set of maps $g^{(\mathrm{enc})}:\mathbb{R}^{s\times d_y}\to[-1,1]^m$ formed by composing L GRU layers and reading out the top-layer final hidden state,

$$g^{(\mathrm{enc})}(H_{t-1}):=h_{L,t-1},\qquad g^{(\mathrm{enc})}=g_L^{(\mathrm{GRU})}\circ\cdots\circ g_1^{(\mathrm{GRU})}, \tag{31}$$

where the layer inputs are chained by $x_{1,i}=y_i$ and $x_{\ell,i}=h_{\ell-1,i}$ for $\ell\in\{2,\dots,L\}$. Since $h_{\ell,t-s-1}=0\in[-1,1]^m$ and each update is the convex combination $h_{\ell,i}=(1-z_{\ell,i})\odot\widetilde{h}_{\ell,i}+z_{\ell,i}\odot h_{\ell,i-1}$ of $h_{\ell,i-1}$ and $\widetilde{h}_{\ell,i}\in(-1,1)^m$, an immediate induction gives $h_{\ell,i}\in[-1,1]^m$, so the output $h_{t-1}\in[-1,1]^m$ as required by (4). In the main text and in the convergence analysis of Section 4, the encoder may be taken to be any $g^{(\mathrm{enc})}\in\mathcal{G}^{(\mathrm{GRU})}_{L,\kappa}(s,m,d_y)$.

A.2. LSTM

With the same dimensions d_y, m, L, s, parameter-norm bound $\kappa > 0$, and input sequence $H_{t-1} = (y_{t-s}, \dots, y_{t-1}) \in \mathbb{R}^{s\times d_y}$ as above, the LSTM encoder differs from the GRU only in the per-layer recursion: each layer maintains both a hidden state $h_{\ell,i} \in \mathbb{R}^m$ and a cell state $c_{\ell,i} \in \mathbb{R}^m$, and uses three gates (input, forget, output) together with a cell candidate.

We define the first LSTM layer class $\mathcal{G}^{(\mathrm{LSTM})}_{1,\kappa}(m, d_y)$ as the set of maps $g^{(\mathrm{LSTM})}_1 : \mathbb{R}^{s\times d_y} \to \mathbb{R}^{s\times m}$, and for $\ell \in \{2, \dots, L\}$ we define the ℓ-th LSTM layer class $\mathcal{G}^{(\mathrm{LSTM})}_{\ell,\kappa}(m)$ as the set of maps $g^{(\mathrm{LSTM})}_\ell : \mathbb{R}^{s\times m} \to \mathbb{R}^{s\times m}$, whose recursion, for $i = t-s, \dots, t-1$, reads

$$i_{\ell,i} = \sigma\left(W^{(i)}_\ell x_{\ell,i} + U^{(i)}_\ell h_{\ell,i-1} + b^{(i)}_\ell\right), \tag{32}$$

$$f_{\ell,i} = \sigma\left(W^{(f)}_\ell x_{\ell,i} + U^{(f)}_\ell h_{\ell,i-1} + b^{(f)}_\ell\right), \tag{33}$$

$$o_{\ell,i} = \sigma\left(W^{(o)}_\ell x_{\ell,i} + U^{(o)}_\ell h_{\ell,i-1} + b^{(o)}_\ell\right), \tag{34}$$

$$\widetilde{c}_{\ell,i} = \tanh\left(W^{(c)}_\ell x_{\ell,i} + U^{(c)}_\ell h_{\ell,i-1} + b^{(c)}_\ell\right), \tag{35}$$

$$c_{\ell,i} = f_{\ell,i} \odot c_{\ell,i-1} + i_{\ell,i} \odot \widetilde{c}_{\ell,i}, \tag{36}$$

$$h_{\ell,i} = o_{\ell,i} \odot \tanh(c_{\ell,i}). \tag{37}$$

Here $i_{\ell,i}, f_{\ell,i}, o_{\ell,i} \in (0,1)^m$ are the input, forget, and output gates, and $\widetilde{c}_{\ell,i} \in (-1,1)^m$ is the cell candidate. The input weight matrices are $W^{(i)}_1, W^{(f)}_1, W^{(o)}_1, W^{(c)}_1 \in \mathbb{R}^{m\times d_y}$ for the first layer and $W^{(i)}_\ell, W^{(f)}_\ell, W^{(o)}_\ell, W^{(c)}_\ell \in \mathbb{R}^{m\times m}$ for $\ell \in \{2, \dots, L\}$; the recurrent weight matrices are $U^{(i)}_\ell, U^{(f)}_\ell, U^{(o)}_\ell, U^{(c)}_\ell \in \mathbb{R}^{m\times m}$; and the bias vectors of the input gate, forget gate, output gate, and cell candidate are $b^{(i)}_\ell, b^{(f)}_\ell, b^{(o)}_\ell, b^{(c)}_\ell \in \mathbb{R}^m$. All entries of these weight matrices and bias vectors are bounded in absolute value by κ. Initial hidden and cell states are taken to be zero, $h_{\ell,t-s-1} = 0$ and $c_{\ell,t-s-1} = 0$.

We define the LSTM encoder function class $\mathcal{G}^{(\mathrm{LSTM})}_{L,\kappa}(s, m, d_y)$ as the set of maps $g^{(\mathrm{enc})} : \mathbb{R}^{s\times d_y} \to [-1,1]^m$ formed by composing L LSTM layers and reading out the top-layer final hidden state,

$$g^{(\mathrm{enc})}(H_{t-1}) := h_{L,t-1}, \qquad g^{(\mathrm{enc})} = g^{(\mathrm{LSTM})}_L \circ \cdots \circ g^{(\mathrm{LSTM})}_1, \tag{38}$$

where the layer inputs are chained by $x_{1,i} = y_i$ and $x_{\ell,i} = h_{\ell-1,i}$ for $\ell \in \{2, \dots, L\}$. Since $o_{\ell,i} \in (0,1)^m$ and $\tanh(c_{\ell,i}) \in (-1,1)^m$, we have $h_{\ell,i} \in (-1,1)^m$ at every layer and time index, so the output $h_{t-1} \in [-1,1]^m$ as required by (4). In the main text and in the convergence analysis of Section 4, the encoder may equivalently be taken to be any $g^{(\mathrm{enc})} \in \mathcal{G}^{(\mathrm{LSTM})}_{L,\kappa}(s, m, d_y)$.

B. Additional experimental details

B.1. Extra results

Figure 5 motivates the choice of grid parameters. For each tick spacing δ, we report the smallest depth d^* whose half-width $d^* \cdot \delta$ exceeds the 95% and 98% quantiles of absolute 5-second mid-price changes. The choice of δ trades off two effects: a larger tick reaches the same coverage with fewer levels, but a smaller tick better resolves consecutive moves, since price changes below δ are aggregated into a single bin. The configuration $(d = 5,\ \delta = 2)$ is a balance of both by attaining the 98% target while keeping the queue dimensionality $d_y = 11$ tractable.

Table 4 stratifies the headline CRPS by realized-volatility regime: evaluation windows are sorted by the standard deviation of mid-price returns over their conditioning context and split into low/mid/high terciles. LOBGrad attains the lowest queue CRPS in every regime and the price-only ablation LG-R the lowest returns CRPS, showing that the gains hold across volatility conditions rather than being driven by a single regime.

Figure 6 shows Pearson correlation matrices between all $2d{=}10$ queue dimensions for the real held-out data and the generated data for LOBGrad and CGAN. LOBGrad was able to learn the block correlation pattern and achieve a lower root mean squared error (RMSE) compared to CGAN.

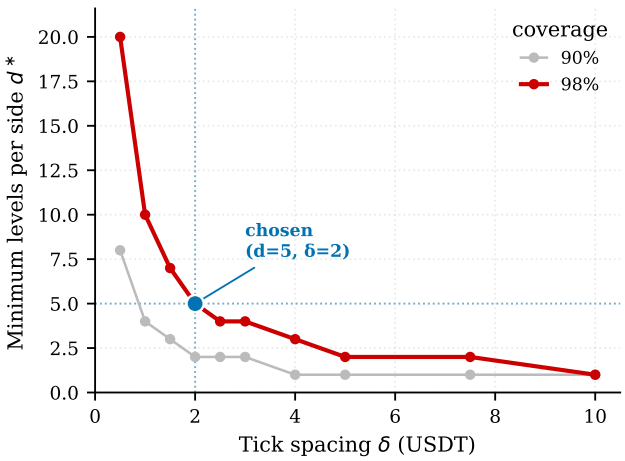


Figure 5. Minimum number of retained levels per side d^* required to cover a target fraction of absolute 5-second BTC/USDT mid-price changes, as a function of tick spacing δ. The chosen configuration (d=5, δ=2) covers ≈98% of observed price moves between consecutive snapshots.

Table 4. Queue and returns CRPS by realized-volatility regime. Lower is better. **Bold** marks the best model per column.

Model	Queue CRPS			Returns CRPS		
	low	mid	high	low	mid	high
LOBGrad	**0.2246**	**0.2547**	**0.3021**	0.8146	1.3150	3.0270
LG-NE	0.2351	0.2701	0.3221	0.8246	1.3611	3.1024
LG-Q	0.2298	0.2553	0.3024	—	—	—
LG-R	—	—	—	**0.8132**	**1.3040**	**2.9799**
CGAN	0.2390	0.2762	0.3377	0.8418	1.3840	3.2975
Hawkes	0.3199	0.3283	0.3747	—	—	—
Poisson	0.2734	0.3022	0.3623	—	—	—

Figure 7 compares marginal queue-size distributions on level 1 and 2 against the held-out data. LOBGrad tracks both more closely than CGAN and the statistical baselines.

B.2. Implementation details

Table 5 indicates the selected hyperparameters of LOBGrad. We compare LOBGrad against three baselines: CGAN, Hawkes, and Poisson. For each baseline, we specify input/output, estimation objective, tuning, conditioning, rollout, and metric applicability.

The conditional WGAN-GP (CGAN) baseline takes as input a context window of joint price–queue features $(\Delta p, \tilde{Q})$ and generates future joint paths over a horizon $H = 20$. It uses a GRU encoder to summarize the context and an autoregressive LSTM generator to produce future trajectories, with latent noise sampled for each generated path. The discriminator is conditioned on the same context and observes the concatenation of the context with either the realized or generated future sequence, and the model is trained with the Wasserstein objective augmented by a gradient penalty. Hyperparameters are selected by Bayesian optimization with Optuna on validation windows, searching over the latent dimension in $[32, 256]$ on a logarithmic scale, the hidden dimension in $[64, 512]$ on a logarithmic scale, one to three encoder and decoder layers, generator and discriminator learning rates in $[10^{-5}, 10^{-3}]$ on a logarithmic scale, $n_{\text{critic}} \in \{1, 2, 3\}$, gradient-penalty weight $\lambda_{\text{GP}} \in [1, 50]$ on a logarithmic scale, stride in $\{5, 10, 20\}$, and queue mode in $\{\texttt{absolute}, \texttt{delta}\}$. Because the model explicitly conditions on the past through the GRU state and generates both queue and price components, all queue and price metrics are applicable.

The Hawkes queue-flow baseline takes the normalized queue history as input, inverse-transforms it to raw queue units for calibration, and generates queue-only future paths without modeling the price process. Its parameters are estimated from

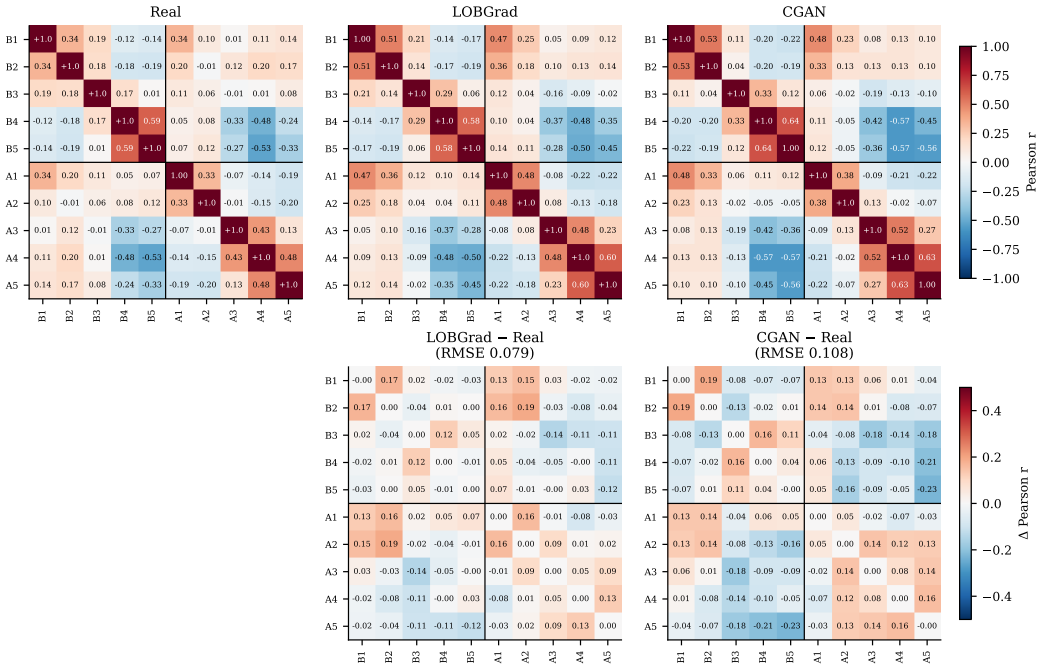


Figure 6. Pearson correlation matrices for LOBGrad and CGAN across $2d = 10$ queue dimensions (top). Difference between correlation real and generated correlation matrix (bottom).

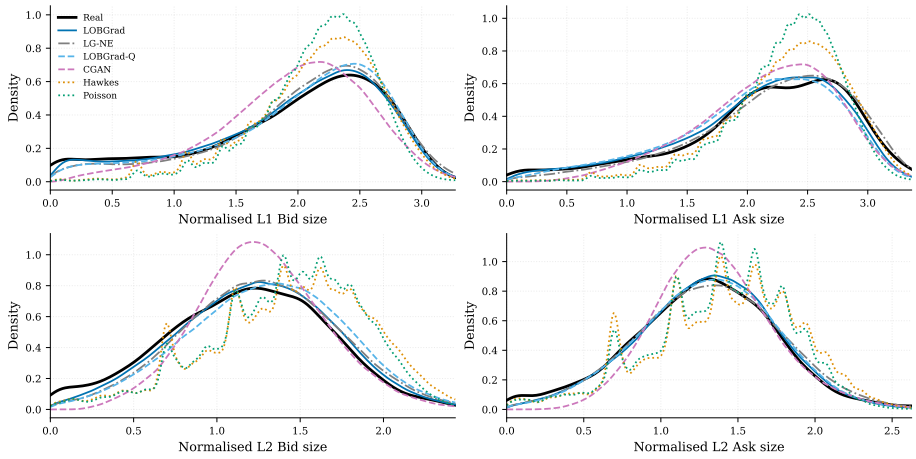


Figure 7. Marginal distribution of level 1 queue (first row) and level 2 queue sizes (second row).

queue differences between successive retained snapshots: positive differences are treated as order submissions, negative differences as removals, and best-level removals are split into market-order and cancellation components using a fixed market fraction of 0.30. Per-level submission and cancellation rates are estimated by moments, while four one-dimensional discrete-time Hawkes intensities, corresponding to market buys, market sells, aggregate ask cancellations, and aggregate bid cancellations, are fitted by Poisson maximum likelihood under the constraints $\mu > 0$, $\alpha \geq 0$, $\beta > 0$, and $\alpha < \beta$, using Nelder–Mead optimization. The resulting objective is the Poisson negative log-likelihood for event-count sequences. We do not perform a hyperparameter search for this baseline and instead use fixed initialization and solver settings, including $\beta_0 = 1$ and `maxiter`=200. Conditioning enters only weakly through the last observed queue state and through intensity warm-starting by replaying the context, and rollout is performed by superposed Poisson event simulation with Hawkes-updated intensities at each step. Since this baseline does not generate a price path, price-return metrics such as return Wasserstein distance, absolute-return ACF, and return CRPS are not applicable.

The Poisson baseline shares the same queue-flow decomposition as Hawkes but uses time-homogeneous rates (no self-

excitation). Submission rates, queue-proportional cancellation rates, and best-level market rates are estimated from training-set empirical moments. Future queue paths are generated by Poisson–multinomial superposition with context initialized at the latest observed queue state. As above, price-return metrics are not applicable.

Table 5. LOBGrad hyperparameters selected by Bayesian optimization (Optuna)

Hyperparameter	Value
Encoder	
Cell type	LSTM
Hidden size m	151
Depth L	1
Look-back window s	100
Dropout	0.178
Noise network	
Type	MLP
Width W	255
Depth	3
Step embedding dim	32
Diffusion	
Steps K	102
Noise schedule	linear
Training	
Learning rate	4.04×10^{-3}
Batch size	64
Epochs	44
Optimizer	Adam

B.3. Evaluation metrics

All generation metrics are computed on 500 test windows sampled without replacement. Each window contains a 100-snapshot conditioning context and a $\tau = 20$-step forecast horizon; each generative model produces 100 autoregressive rollouts per window. Confidence intervals in Table 1 are obtained by bootstrapping evaluation windows with 1000 resamples.

We follow LOB-Bench (Nagy et al., 2025) and report Wasserstein-1 distances for three marginal quantities: best-level total queue volume, order-book imbalance, and the return distribution. We additionally report four dependence and profile diagnostics. Queue-profile MAE compares the average generated and realized depth profiles across levels and forecast steps, measuring whether the model reproduces the empirical decay of liquidity away from the mid-price (Bouchaud et al., 2018). Queue ACF MAE computes the MAE between generated and realized autocorrelation vectors of the top-of-book total queue size at lags $1, \ldots, 5$. Absolute-return ACF MAE applies the same construction to absolute returns, capturing short-horizon volatility clustering (Cont, 2001).

For probabilistic calibration, we use the continuous ranked probability score. For a predictive distribution F and realization y,

$$\mathrm{CRPS}(F, y) = \int_{-\infty}^{\infty} \big(F(z) - \mathbf{1}\{z \geq y\}\big)^2 \, \mathrm{d}z. \tag{39}$$

CRPS is a strictly proper scoring rule, uniquely minimized by the true conditional distribution (Gneiting & Raftery, 2007). In our experiments, F is approximated by the empirical distribution of the 100 rollouts, and CRPS is averaged over forecast steps, windows, and relevant dimensions.

Finally, to provide direct evidence for probabilistic calibration beyond CRPS, we evaluate central prediction-interval coverage on the same 500 test windows, each with a 100-snapshot context and a $\tau = 20$-step horizon. For each window origin w, horizon step h, and variable dimension j, we draw $M = 500$ autoregressive rollouts $\{\hat{y}^{(m)}_{w,h,j}\}_{m=1}^{M}$. For a nominal central level $c \in \{0.5, 0.8, 0.9, 0.95\}$, we form the empirical central interval $[\hat{q}_{(1-c)/2,w,h,j},\ \hat{q}_{(1+c)/2,w,h,j}]$, where $\hat{q}_{\alpha,w,h,j}$

is the empirical α-quantile of the M rollouts. For a variable group G, empirical coverage is

$$\widehat{\mathrm{Cov}}_{c,h,G} = \frac{1}{W|G|} \sum_{w=1}^{W} \sum_{j \in G} \mathbf{1}\big\{ \hat{q}_{(1-c)/2,w,h,j} \le y_{w,h,j} \le \hat{q}_{(1+c)/2,w,h,j} \big\} .$$

We also report interval sharpness via the mean width

$$\widehat{\mathrm{Width}}_{c,G} = \frac{1}{WH|G|} \sum_{w=1}^{W} \sum_{h=1}^{H} \sum_{j \in G} \big(\hat{q}_{(1+c)/2,w,h,j} - \hat{q}_{(1-c)/2,w,h,j} \big) .$$

We report four groups: returns, queue dimensions (10d), best-level total queue volume, and order-book imbalance.

C. Proofs for Section 4

This appendix proves the excess-risk bounds in Section 4. The analysis is for the DDPM ε-prediction loss. The transformed LOB vector is assumed bounded, as in Assumption 4.1; the only unbounded random variable is the Gaussian noise injected by the forward DDPM. Constants denoted by $C, c, C_1, \dots$ may depend on fixed quantities such as s, d_y, m, K, B_y, the diffusion schedule, and the mixing constants, but not on T, V, W unless stated otherwise. The parameter bound κ is taken large enough to contain the approximating networks; any polynomial dependence of κ on the localization radius contributes only logarithmic factors and is absorbed into $\widetilde{O}$.

For $r \ge 1$, define the Gaussian localization event and domains

$$E_t(r) := \{ \|\varepsilon_t\|_\infty \le r \}, \qquad B_r := B_y + r, \tag{40}$$

$$\Omega_r := ([-B_y, B_y]^{d_y})^s \times [-B_r, B_r]^{d_y} \times \{1, \dots, K\}, \qquad \Xi_r := [-B_r, B_r]^{d_y} \times [-1,1]^m. \tag{41}$$

On $E_t(r)$, $\|y_t^{(N_t)}\|_\infty \le B_y + r = B_r$, and hence $(H_{t-1}, y_t^{(N_t)}, N_t) \in \Omega_r$.

The localized head and encoder approximation errors are

$$A^{\mathrm{head}}_{D,W}(r) := \inf_{g^{(\epsilon)} \in \mathcal{G}^{(\epsilon)}_{D,W,\kappa}(d_y,m,K)} \max_{1 \le k \le K} \big\| g^{(\epsilon)}(\cdot,k,\cdot) - \bar{g}^\star(\cdot,k,\cdot) \big\|^2_{\infty,\Xi_r}, \tag{42}$$

$$\Delta_{\mathrm{enc}}(V,L) := \inf_{g^{(\mathrm{enc})} \in \mathcal{G}^{(\mathrm{enc})}_{V,L,\kappa}(s,m,d_y)} \sup_{H \in ([-B_y,B_y]^{d_y})^s} \big\| g^{(\mathrm{enc})}(H) - \Gamma^\dagger(H) \big\|_\infty . \tag{43}$$

The main text uses $A^{\mathrm{head}}(D,W) = A^{\mathrm{head}}_{D,W}(r_T)$ with $r_T \asymp \sqrt{\log T}$.

C.1. Bayes identity and model assumptions

Let $X_t = (H_{t-1}, y_t^{(N_t)}, N_t)$ and $g^\star(X_t) = \mathbb{E}[\varepsilon_t \mid X_t]$. For every measurable predictor g with finite risk,

$$\mathcal{R}(g) - \mathcal{R}(g^\star) = \mathbb{E}\big[\|g(X_t) - g^\star(X_t)\|_2^2 \big]. \tag{44}$$

Indeed, expanding the square gives

$$\begin{aligned} \mathcal{R}(g) &= \mathbb{E}\big[\|\varepsilon_t - g^\star(X_t) + g^\star(X_t) - g(X_t)\|_2^2 \big] \\ &= \mathcal{R}(g^\star) + \mathbb{E}\big[\|g(X_t) - g^\star(X_t)\|_2^2 \big] + 2\mathbb{E}\big[(\varepsilon_t - g^\star(X_t))^\top (g^\star(X_t) - g(X_t)) \big], \end{aligned}$$

and the last term is zero by iterated expectation.

Assumption C.1 (Bounded support, mixing, and auxiliary randomness)**.** The process $(y_t)_{t \in \mathbb{Z}}$ is strictly stationary, exponentially α-mixing, and satisfies $\|y_t\|_\infty \le B_y$ almost surely. The auxiliary variables (N_t, ε_t) are i.i.d. across t, independent of the LOB process, with $N_t \sim \mathrm{Unif}\{1, \dots, K\}$ and $\varepsilon_t \sim \mathcal{N}(0, I_{d_y})$.

Assumption C.2 (Factorization and smoothness)**.** There exist maps $\Gamma^\dagger : ([-B_y, B_y]^{d_y})^s \to [-1,1]^m$ and $\bar{g}^\star : \mathbb{R}^{d_y} \times \{1, \dots, K\} \times [-1,1]^m \to \mathbb{R}^{d_y}$ such that

$$g^\star(H, y, k) = \bar{g}^\star\big(y, k, \Gamma^\dagger(H)\big). \tag{45}$$

The map $\Gamma^\dagger$ is γ-Hölder on $([-B_y, B_y]^{d_y})^s$. For every $r \ge 1$ and k, each component of $(y,h) \mapsto \bar{g}^\star(y,k,h)$ belongs to a β-Hölder ball on Ξ_r with radius at most $C_g(1+r)^{\rho_g}$.

The oracle has a linear envelope. Let $a_k = \sqrt{\bar{\alpha}_k}$, $\sigma_k = \sqrt{1-\bar{\alpha}_k}$, and $\sigma_{\min} = \min_{1\le k\le K} \sigma_k > 0$. Since $y^{(k)} = a_k y + \sigma_k \varepsilon$ and $\|y\|_\infty \le B_y$,

$$\|g^\star(H,z,k)\|_\infty = \left\| \mathbb{E}\left[\frac{z - a_k y}{\sigma_k} \,\middle|\, H, z, k \right] \right\|_\infty \le \sigma_{\min}^{-1}(\|z\|_\infty + B_y) \le C_\star(1 + \|z\|_\infty). \tag{46}$$

We choose $C_{\mathrm{clip}} > C_\star$ in (6), so the oracle is not altered by the outer clipping.

Assumption C.3 (Entropy and envelope of the clipped class)**.** Let $\mathcal{G} = \mathcal{G}^{(\mathrm{tg})}_{V,L,D,W,\kappa}(s, m, d_y, K)$. For every $r \ge 1$ and $0 < \eta \le 1$,

$$\log N\big(\eta, \mathcal{G}, \|\cdot\|_{\infty,\Omega_r}\big) \le C_{\mathrm{ent}} \mathfrak{S}(V,L,D,W) \log\left(\frac{C_{\mathrm{ent}}(1+r)^{C_{\mathrm{ent}}}}{\eta} \right). \tag{47}$$

Moreover, every $g \in \mathcal{G}$ satisfies

$$\|g(H,y,k)\|_\infty \le C_{\mathrm{ent}}(1 + \|y\|_\infty). \tag{48}$$

C.2. Empirical-process bound

Lemma C.4 (Finite-class deviation under exponential mixing)**.** *Suppose Assumption C.1 holds and the original LOB process satisfies $\alpha_y(q) \le C_\alpha e^{-c_\alpha q}$. For fixed look-back length s, the augmented process $Z_t = (H_{t-1}, y_t^{(N_t)}, N_t, \varepsilon_t)$ is exponentially α-mixing. There exist constants $C_Z, c_Z > 0$, depending only on s, C_α, c_α, such that $\alpha_Z(q) \le C_Z e^{-c_Z q}$ for $q \ge 1$. Let $\mathcal{H}$ be a finite class of measurable functions of Z_t such that $0 \le h(Z_t) \le B_h$ almost surely for all $h \in \mathcal{H}$. Then, for every $x > 0$, with probability at least $1 - e^{-x}$,*

$$\sup_{h\in\mathcal{H}} \left| \frac{1}{T}\sum_{t=1}^{T} h(Z_t) - \mathbb{E}[h(Z_t)] \right| \le C B_h \left[\sqrt{\frac{(\log|\mathcal{H}| + x)\log T}{T}} + \frac{(\log|\mathcal{H}| + x)(\log T)^2}{T} \right]. \tag{49}$$

Proof. Let $U_t = (N_t, \varepsilon_t)$. Since $(U_t)_{t\in\mathbb{Z}}$ is i.i.d. and independent of the LOB process, adjoining U_t does not increase long-range dependence except through the overlap of the history windows. More precisely, for $q > s$, $\alpha_Z(q) \le \alpha_y(q-s)$, because Z_t is measurable with respect to $\sigma(y_{t-s}, \dots, y_t, U_t)$. Hence, if $\alpha_y(q) \le C_\alpha e^{-c_\alpha q}$, then $\alpha_Z(q) \le C_Z e^{-c_Z q}$ for constants $C_Z, c_Z > 0$, with the constants depending on the fixed look-back length s.

Fix $h \in \mathcal{H}$ and define

$$X_t^h = h(Z_t) - \mathbb{E}[h(Z_t)], \qquad S_T^h = \sum_{t=1}^{T} X_t^h.$$

Then $\mathbb{E}[X_t^h] = 0$, $\|X_t^h\|_\infty \le B_h$, and $(X_t^h)_{t\ge1}$ is exponentially α-mixing with the same exponential rate, up to constants. Let

$$v_h^2 = \sup_{i\ge1} \left\{ \mathrm{Var}(X_i^h) + 2\sum_{j>i} |\operatorname{Cov}(X_i^h, X_j^h)| \right\}.$$

By the covariance inequality for bounded strongly mixing variables,

$$|\operatorname{Cov}(X_i^h, X_j^h)| \le 4B_h^2 \alpha_Z(j-i),$$

and therefore $v_h^2 \le C B_h^2$, since $\sum_{\ell\ge1} \alpha_Z(\ell) < \infty$.

Applying Theorem 2 of (Merlevède et al., 2009) to $(X_t^h)_{t=1}^T$, we obtain constants $c, C > 0$ such that, for all $u > 0$,

$$\mathbb{P}\left(|S_T^h| \ge u\right) \le 2\exp\left[-c\, \frac{u^2}{TB_h^2 + B_h u (\log T)^2} \right].$$

Equivalently,

$$\mathbb{P}\left(\left| \frac{1}{T} S_T^h \right| \ge C B_h \left[\sqrt{\frac{a}{T}} + \frac{a(\log T)^2}{T} \right] \right) \le 2e^{-a}$$

for every $a > 0$, after changing C.

Now take $a = x + \log(2|\mathcal{H}|)$ and apply a union bound over $h \in \mathcal{H}$. Since $\sqrt{a/T} \le \sqrt{a \log T / T}$ for $T \ge e$, and constants absorb the factor $\log 2$, the displayed bound follows. □

For $g \in \mathcal{G}$, write

$$\ell_g(Z_t) = \|\varepsilon_t - g(H_{t-1}, y_t^{(N_t)}, N_t)\|_2^2, \qquad \ell_g^{(r)}(Z_t) = \ell_g(Z_t)\mathbf{1}_{E_t(r)}.$$

Lemma C.5 (Localized loss entropy and Gaussian remainder)**.** *Under Assumptions C.1–C.3, the class* $\mathcal{L}_r = \{\ell_g^{(r)} : g \in \mathcal{G}\}$ *satisfies, for* $0 < \eta \le 1$*,*

$$\log N\big(\eta, \mathcal{L}_r, \|\cdot\|_\infty\big) \le C\mathfrak{S}(V, L, D, W) \log\left(\frac{C(1+r)^C}{\eta}\right), \tag{50}$$

and $0 \le \ell_g^{(r)} \le B_\ell(r) := Cd_y(1+r)^2$*. Moreover,*

$$\mathbb{P}\left(\bigcap_{t=1}^{T} E_t(r)\right) \ge 1 - CTd_y e^{-cr^2}, \tag{51}$$

$$\sup_{g\in\mathcal{G}} \mathbb{E}\big[\ell_g(Z_t)\mathbf{1}_{E_t(r)^c}\big] \le C(1+r)^C e^{-cr^2}, \tag{52}$$

$$\sup_{g\in\mathcal{G}} \mathbb{E}\big[\|g(X_t) - g^\star(X_t)\|_2^2 \mathbf{1}_{E_t(r)^c}\big] \le C(1+r)^C e^{-cr^2}. \tag{53}$$

Proof. On $E_t(r)$, $(H_{t-1}, y_t^{(N_t)}, N_t) \in \Omega_r$ and $\|\varepsilon_t\|_\infty \le r$. The envelope (48) gives $\|g(H_{t-1}, y_t^{(N_t)}, N_t)\|_\infty \le C(1+r)$, so $\ell_g^{(r)} \le Cd_y(1+r)^2$. Also, for $g, g' \in \mathcal{G}$,

$$|\ell_g - \ell_{g'}| \le Cd_y(1+r)\|g - g'\|_{\infty,\Omega_r} \quad \text{on } E_t(r).$$

Thus an appropriately rescaled cover of $\mathcal{G}$ on Ω_r induces a cover of $\mathcal{L}_r$, proving (50).

The event probability follows from a union bound over the d_y Gaussian coordinates. For the remainders, use the clipped-class envelope and (46) together with $\|y_t^{(N_t)}\|_\infty \le B_y + \|\varepsilon_t\|_\infty$. Both ℓ_g and $\|g - g^\star\|_2^2$ are then bounded by a fixed polynomial in $1 + \|\varepsilon_t\|_\infty$. Gaussian tail integration gives (52)–(53). □

Theorem C.6 (Localized estimation)**.** *Suppose Assumptions C.1 and C.3 hold. For every* $r \ge 1$ *and* $\delta \in (0,1)$*, with probability at least* $1 - \delta - CTd_y e^{-cr^2}$*,*

$$\begin{aligned}\mathcal{R}(\widehat{g}_T^{(\mathrm{tg})}) - \inf_{g\in\mathcal{G}} \mathcal{R}(g) \le CB_\ell(r)\left[\sqrt{\frac{v_{T,r}(\delta)\log T}{T}} + \frac{v_{T,r}(\delta)(\log T)^2}{T}\right] \\ + \frac{C}{T} + C(1+r)^C e^{-cr^2},\end{aligned} \tag{54}$$

where

$$v_{T,r}(\delta) := C\mathfrak{S}(V, L, D, W)\log\big(CT(1+r)^C\big) + \log(1/\delta). \tag{55}$$

Proof. Let $E_T(r) = \cap_{t=1}^T E_t(r)$. On $E_T(r)$, $\mathcal{R}_T(g) = T^{-1}\sum_t \ell_g^{(r)}(Z_t)$ for all $g \in \mathcal{G}$. By Lemma C.5, choose a $1/T$-net of $\mathcal{L}_r$ whose logarithmic size is bounded by the first term in $v_{T,r}(\delta)$. Applying Lemma C.4 to this net gives a uniform deviation bound for the net; the $1/T$ approximation adds at most C/T. The Gaussian remainder in Lemma C.5 converts the truncated deviation to a deviation for $\mathcal{R}_T - \mathcal{R}$. Thus, on the stated event,

$$\sup_{g\in\mathcal{G}} |\mathcal{R}_T(g) - \mathcal{R}(g)| \le CB_\ell(r)\left[\sqrt{\frac{v_{T,r}(\delta)\log T}{T}} + \frac{v_{T,r}(\delta)(\log T)^2}{T}\right] + \frac{C}{T} + C(1+r)^C e^{-cr^2}.$$

The empirical minimizer inequality $\mathcal{R}(\widehat{g}_T^{(\mathrm{tg})}) \le \mathcal{R}(g) + 2\sup_{f\in\mathcal{G}} |\mathcal{R}_T(f) - \mathcal{R}(f)|$ for all $g \in \mathcal{G}$ completes the proof. □

Define the localized composite approximation error

$$A_{V,L,D,W}(r) := \inf_{g \in \mathcal{G}^{(\mathrm{tg})}_{V,L,D,W,\kappa}(s,m,d_y,K)} \|g - g^\star\|_{\infty,\Omega_r}^2. \tag{56}$$

Combining Theorem C.6 with the Bayes identity gives

$$\mathcal{R}(\widehat{g}_T^{(\mathrm{tg})}) - \mathcal{R}(g^\star) \le C A_{V,L,D,W}(r) + C B_\ell(r) \left[\sqrt{\frac{v_{T,r}(\delta)\log T}{T}} + \frac{v_{T,r}(\delta)(\log T)^2}{T} \right] + \frac{C}{T} + C(1+r)^C e^{-cr^2}. \quad (57)$$

Indeed, on $E_t(r)$ the approximation part is bounded by $d_y \|g - g^\star\|^2_{\infty,\Omega_r}$, and on $E_t(r)^c$ it is bounded by (53).

C.3. Approximation bounds

Theorem C.7 (Head–encoder decomposition). *Under Assumptions C.2–C.3, with* $\beta_0 = \min\{\beta, 1\}$,

$$A_{V,L,D,W}(r) \le C\Big\{ A^{\mathrm{head}}_{D,W}(r) + (1+r)^C \Delta_{\mathrm{enc}}(V,L)^{2\beta_0} \Big\}. \quad (58)$$

Proof. Pick $g_\eta^{(\epsilon)}$ and $g_\eta^{(\mathrm{enc})}$ within η of the infima in (42)–(43), and form the clipped composition $\widetilde{g}_\eta^{(\mathrm{tg})}$. For $(H,y,k) \in \Omega_r$, clipping is 1-Lipschitz in the second argument and leaves $g^\star$ unchanged by (46). Therefore

$$\begin{aligned} \|\widetilde{g}_\eta^{(\mathrm{tg})}(H,y,k) - g^\star(H,y,k)\|_\infty &\le \|g_\eta^{(\epsilon)}(y,k,g_\eta^{(\mathrm{enc})}(H)) - \bar{g}^\star(y,k,g_\eta^{(\mathrm{enc})}(H))\|_\infty \\ &\quad + \|\bar{g}^\star(y,k,g_\eta^{(\mathrm{enc})}(H)) - \bar{g}^\star(y,k,\Gamma^\dagger(H))\|_\infty. \end{aligned}$$

The first term is at most $(A^{\mathrm{head}}_{D,W}(r) + \eta)^{1/2}$. The second is at most $C(1+r)^C(\Delta_{\mathrm{enc}}(V,L) + \eta)^{\beta_0}$ by the local Hölder property of $\bar{g}^\star$. Squaring and letting $\eta \downarrow 0$ proves (58). □

Proposition C.8 (Head approximation). *If K is fixed and $D(W) = O(\log W)$, then*

$$A^{\mathrm{head}}_{D(W),W}(r) \le C(1+r)^C W^{-2\beta/(d_y+m)} (\log W)^C. \quad (59)$$

Proof. Let $q = d_y + m$. Fix a diffusion step $k \in \{1,\dots,K\}$ and one output coordinate $j \in \{1,\dots,d_y\}$. After an affine rescaling of $\Xi_r = [-B_y - r, B_y + r]^{d_y} \times [-1,1]^m$ to $[0,1]^q$, the coordinate map $(y,h) \mapsto \bar{g}_j^\star(y,k,h)$ belongs to a β-Hölder ball on $[0,1]^q$ with radius at most $C(1+r)^C$. By the Hölder approximation theorem for sparse deep ReLU networks, specifically Theorem 5 of (Schmidt-Hieber, 2020), for every integer N there exists a ReLU network with width CN, depth $C\log N$, and bounded weights such that

$$\sup_{(y,h)\in\Xi_r} \big|\widehat{g}_{j,k}(y,h) - \bar{g}_j^\star(y,k,h)\big| \le C(1+r)^C N^{-\beta/q}.$$

Here the factor $(1+r)^C$ comes from the rescaling of the domain and from the assumed polynomial growth of the local Hölder radius.

Parallelizing the d_y coordinate networks gives a d_y-output network with the same rate and only a constant-factor increase in width, since d_y is fixed. Since K is fixed, the K diffusion-step slices can also be combined with only a constant-factor overhead. Equivalently, one may view $\mathcal{G}^{(\epsilon)}_{D,W,\kappa}$ as a K-tuple of ReLU heads indexed by the discrete diffusion step k. This is equivalent, up to constants, to using a single ReLU head with a fixed one-hot or piecewise-linear step encoding.

Taking $N \asymp W$ and $D(W) = C\log W$ yields

$$\max_{1\le k\le K} \|\widehat{g}(\cdot,k,\cdot) - \bar{g}^\star(\cdot,k,\cdot)\|_{\infty,\Xi_r} \le C(1+r)^C W^{-\beta/(d_y+m)} (\log W)^C.$$

Squaring the last display gives

$$A^{\mathrm{head}}_{D(W),W}(r) \le C(1+r)^C W^{-2\beta/(d_y+m)} (\log W)^C.$$

□

Proposition C.9 (Generic encoder approximation). *For fixed s, d_y, m, γ, there exists a recurrent depth $L_0 = L_0(s,d_y,m,\gamma)$, independent of V and T, such that*

$$\Delta_{\mathrm{enc}}(V,L_0) \le C V^{-2\gamma/(sd_y)} (\log V)^C, \qquad \Delta_{\mathrm{enc}}(V,L_0)^{2\beta_0} \le C V^{-4\gamma\beta_0/(sd_y)} (\log V)^C. \quad (60)$$

Proof. After rescaling $([-B_y, B_y]^{d_y})^s$ to a unit cube, $\Gamma^\dagger$ is an m-output γ-Hölder map on dimension sd_y. The recurrent-network approximation theorem of (Jiao et al., 2024) gives a fixed-depth recurrent ReLU encoder with width $CJ \log J$ and error $CJ^{-2\gamma/(sd_y)}$. Setting $J \asymp V/\log V$ proves the first bound; the second follows by raising to $2\beta_0$. □

Proof of Theorem 4.3. Choose $r_T = c_r\sqrt{\log T}$ so that $Te^{-cr_T^2} = O(T^{-2})$. Apply (57) with $r = r_T$ and then use Theorem C.7. The factors polynomial in r_T and all logarithmic terms are absorbed into $\widetilde{O}_{\mathbb{P}}$, giving (17). The complexity reduction $\mathfrak{S}(V, L, D, W) = \widetilde{O}(V^2 + W^2)$ for fixed L and $D = O(\log W)$ follows from Proposition C.11 below. □

Proof of Corollary 4.5. By Propositions C.8 and C.9,

$$A^{\mathrm{head}}_{D(W),W} = \widetilde{O}(W^{-2\beta/(d_y+m)}), \qquad \Delta_{\mathrm{enc}}(V, L_0)^{2\beta_0} = \widetilde{O}(V^{-4\gamma\beta_0/(sd_y)}).$$

Together with $\mathfrak{S}(V, L_0, D(W), W) = \widetilde{O}(V^2 + W^2)$, Theorem 4.3 yields

$$\mathcal{R}(\widehat{g}_T^{(\mathrm{tg})}) - \mathcal{R}(g^\star) = \widetilde{O}_{\mathbb{P}}\left(\sqrt{\frac{V^2+W^2}{T}} + \frac{V^2+W^2}{T} + W^{-2\beta/(d_y+m)} + V^{-4\gamma\beta_0/(sd_y)}\right).$$

Balancing $W/\sqrt{T}$ with $W^{-2\beta/(d_y+m)}$ and $V/\sqrt{T}$ with $V^{-4\gamma\beta_0/(sd_y)}$ gives (19) and (20); the second-order term is then smaller. □

C.4. Entropy verification

Lemma C.10 (Covering bounded-parameter ReLU networks)**.** *Let $\mathcal{H}(d_{\mathrm{in}}, q, \bar{W}, \bar{D}, \bar{\kappa})$ be the class of q-output ReLU feedforward networks with input dimension d_{in}, hidden widths at most $\bar{W}$, depth at most $\bar{D}$, and affine parameters bounded by $\bar{\kappa}$. For $B \geq 1$ and $0 < \eta \leq 1$,*

$$\log N\big(\eta, \mathcal{H}, \|\cdot\|_{\infty,[-B,B]^{d_{\mathrm{in}}}}\big) \leq C\bar{W}^2\bar{D}\log\left(\frac{C(1+B)(\bar{W}\bar{D})^{\bar{D}}}{\eta}\right). \tag{61}$$

Proof. For a fixed architecture, quantize each scalar parameter on a mesh of size proportional to $\eta\{\bar{D}\bar{W}(1+B)(2\bar{\kappa})^{\bar{D}}\}^{-1}$. A layer-by-layer Lipschitz induction shows that the resulting network perturbation is at most η in sup norm on $[-B, B]^{d_{\mathrm{in}}}$. The number of scalar parameters is $O(\bar{W}^2\bar{D})$, and the number of admissible width/depth patterns is absorbed by the logarithmic factor, proving (61). □

Proposition C.11 (Entropy of the clipped LOBGrad class)**.** *For $\mathcal{G}^{(\mathrm{tg})}_{V,L,D,W,\kappa}(s, m, d_y, K)$,*

$$\log N\big(\eta, \mathcal{G}^{(\mathrm{tg})}_{V,L,D,W,\kappa}, \|\cdot\|_{\infty,\Omega_r}\big) \leq C\mathfrak{S}(V, L, D, W)\log\left(\frac{C(1+r)}{\eta}\right), \tag{62}$$

where

$$\mathfrak{S}(V, L, D, W) = \big((2s-1)V + W + d_y + 1\big)^2(sL + D + 2)^2\log\big((V+W)(L+D+2)\big). \tag{63}$$

The envelope (48) holds. If L is fixed and $D = O(\log W)$, then $\mathfrak{S}(V, L, D, W) = \widetilde{O}(V^2 + W^2)$.

Proof. The envelope follows from the outer clipping in (6). For entropy, unroll the recurrent encoder over the s tokens. A width-V, depth-L recurrent encoder is represented as a feedforward ReLU network on $\mathrm{vec}(H) \in \mathbb{R}^{sd_y}$ with width at most $(2s-1)V$ and depth at most sL; see (Jiao et al., 2024). After concatenating $(y, k/K)$ and composing with the denoising head, each composite predictor is a clipped ReLU FNN with width $\bar{W} = (2s-1)V + W + d_y + 1$ and depth $\bar{D} = sL + D + 2$ on a cube of radius $C(1+r)$. Lemma C.10 and the 1-Lipschitz property of clipping imply (62). The displayed $\widetilde{O}(V^2 + W^2)$ bound follows by substituting fixed L and $D = O(\log W)$. □

C.5. Latent-state refinement

For a cell map $\Phi \in \mathcal{M}(L, V, \kappa; m, m)$, define the state-space encoder

$$h_0^\Phi = 0, \qquad h_j^\Phi = \Pi_{[-1,1]^m}\big(\Phi(P^\dagger u_j + h_{j-1}^\Phi)\big), \quad j = 1, \dots, s, \qquad g_\Phi^{(\mathrm{enc,st})}(H) = h_s^\Phi. \tag{64}$$

Let $\mathcal{G}^{(\mathrm{tg,st})}_{V,L,D,W,\kappa}$ be the corresponding LOBGrad class using this encoder and define

$$\Delta_{\mathrm{st}}(V, L) := \inf_{g^{(\mathrm{enc,st})}} \sup_{H \in ([-B_y, B_y]^{d_y})^s} \|g^{(\mathrm{enc,st})}(H) - \Gamma^\dagger(H)\|_\infty. \tag{65}$$

Proposition C.12 (State-space approximation)**.** *Under the contractive realization* (21)*, there is a fixed depth L_1 such that*

$$\Delta_{\mathrm{st}}(V, L_1) \le CV^{-2\gamma/m}(\log V)^C, \qquad \Delta_{\mathrm{st}}(V, L_1)^{2\beta_0} \le CV^{-4\gamma\beta_0/m}(\log V)^C. \tag{66}$$

Proof. Approximate the m-dimensional transition map $\Phi^\dagger$ by a ReLU cell Φ_V with error $\delta_V \le CV^{-2\gamma/m}(\log V)^C$ on the compact set visited by $P^\dagger u + h$, where $\|u\|_\infty \le B_y$ and $h \in [-1, 1]^m$. Let h_j and $h_j^\dagger$ be the approximate and oracle states. Since clipping is 1-Lipschitz and the oracle transition is contractive,

$$\|h_j - h_j^\dagger\|_\infty \le \delta_V + \lambda\|h_{j-1} - h_{j-1}^\dagger\|_\infty.$$

Iterating gives $\|h_s - h_s^\dagger\|_\infty \le \delta_V/(1-\lambda)$, proving the first bound; the second follows by raising to $2\beta_0$. □

Proposition C.13 (Entropy of the state-space class)**.** *The state-space class $\mathcal{G}^{(\mathrm{tg,st})}_{V,L,D,W,\kappa}$ satisfies the same envelope condition as* (48)*. Its metric entropy has the same form as* (62)*; in particular, for fixed L and $D = O(\log W)$,*

$$\mathfrak{S}_{\mathrm{st}}(V, L, D, W) = \widetilde{O}(V^2 + W^2). \tag{67}$$

Proof. The envelope again follows from the outer clipping. For entropy, first apply the fixed affine lift $u \mapsto P^\dagger u$. The learnable recurrent cell then acts on an m-dimensional sequence and can be unrolled over s steps exactly as in Proposition C.11; composing with the same denoising head and clipping gives the stated bound. □

Proof of Corollary 4.6. The proof of Theorem 4.3 applies to the state-space subclass by Proposition C.13. The head approximation is unchanged, while Proposition C.12 replaces the generic encoder term by $\Delta_{\mathrm{st}}(V, L_1)^{2\beta_0} = \widetilde{O}(V^{-4\gamma\beta_0/m})$. Thus

$$\mathcal{R}(\widehat{g}^{(\mathrm{tg})}_{T,\mathrm{st}}) - \mathcal{R}(g^\star) = \widetilde{O}_{\mathbb{P}}\left(\sqrt{\frac{V^2 + W^2}{T}} + \frac{V^2 + W^2}{T} + W^{-2\beta/(d_y+m)} + V^{-4\gamma\beta_0/m}\right).$$

Balancing the two stochastic terms with the two approximation terms yields (23). □